

Linear Integrated Circuit

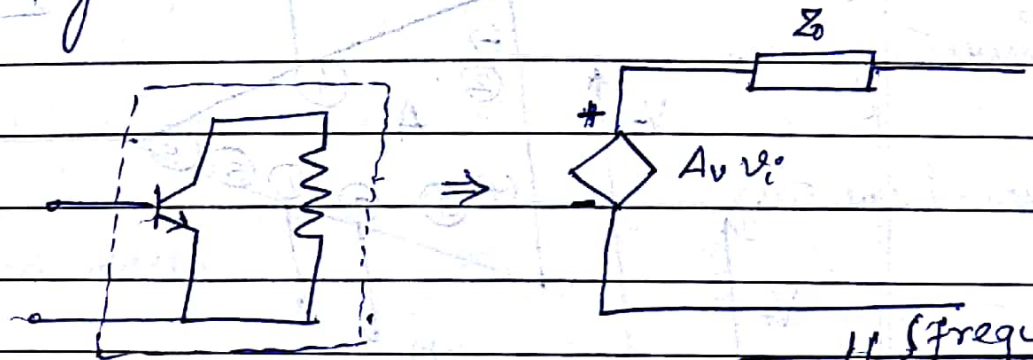
Integrated Circuit Implementation of VCVS

741 $\rightarrow$ first internally compensated IC voltage amplifiers (1966-1967, Fairchild, National Semiconductor, Analog Devices & David Fullagar).

Fairchild $\rightarrow$ $\mu A 741$

National Semiconductors $\rightarrow$ LM741

Analog Devices $\rightarrow$ AD741



first 2 blocks $\rightarrow$

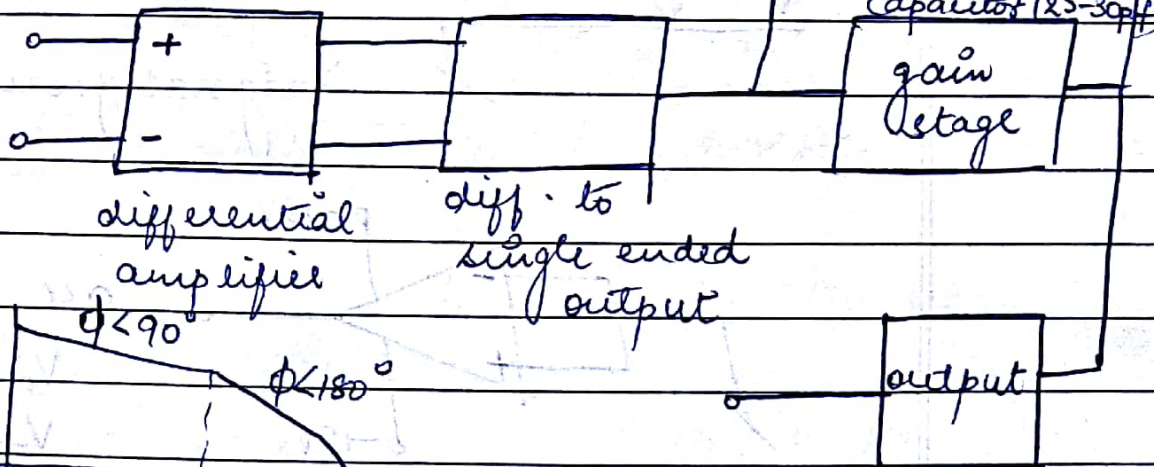
1 pole ;

second block $\rightarrow$

1 pole ;

third block $\rightarrow$

1 pole -



differential amplifier

diff. to single ended output

Frequency Compensation Capacitor (25-30pF)

gain stage

output

$\phi < 90^\circ$

$\phi < 180^\circ$

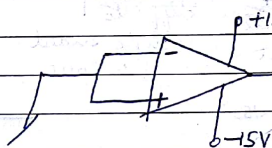
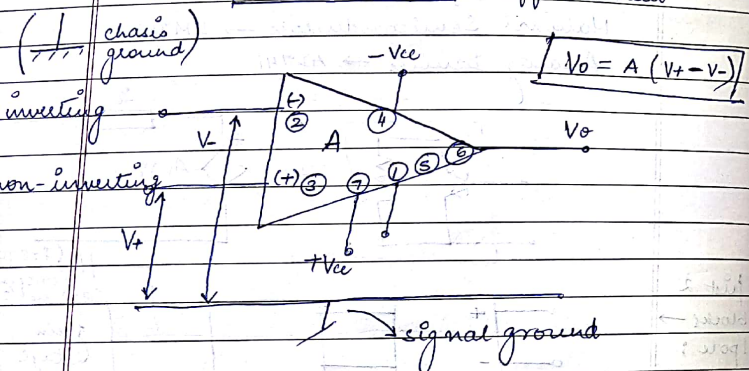
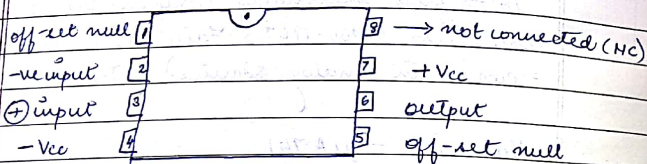
$\phi = 180^\circ$ somewhere

$\Rightarrow$ oscillations in amplifiers $\Rightarrow$ this capacitor used

1 MHz

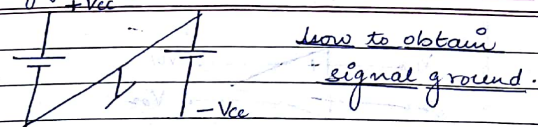
$\rightarrow$ due to capacitor

IC - 741 (pin diagram).
DIP (dual-in-line package)



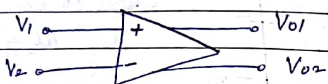
acc to formula,
 $V_o = 0$ as
 $V_+ = V_- = 0$
But $V_o = +15V / -15V$
coz of high gain value
(open loop gain)

frequency at which gain
(i) 3dB (4-5Hz) $\rightarrow 1/\sqrt{2}$ of peak value
(ii) unity gain (1MHz) becomes 1



attributes of good amplifier -

	Ideal	741	Real
① open loop gain	∞		100-120dB (10^5-10^6) typically 2×10^5
② input impedance	∞		4-5M Ω
③ output impedance	0		50-70 Ω
④ bandwidth	∞		(i) 3dB (4-5Hz) (ii) unity gain (1MHz)
⑤ slew. rate $\left\{ \frac{dV_o}{dt} \right\}_{max}$	∞		0.5V/ μ s. if i/p is diff. amplifier, thus lower slew rates
⑥ off set voltage	0		10^{-7} to 10^{-6} V
⑦ bias current input bias current $= (I_+ + I_-) / 2$	0		70-100 μ A
⑧ off-set current ($I_+ - I_-$)	0		30-40 μ A
⑨ CMRR (common mode rejection ratio) $ A_{dm}/A_{cm} $			
⑩ PSRR (power supply rejection ratio)			



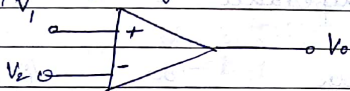
$$V_{id} = V_1 - V_2$$

$$V_{iav} = \frac{V_1 + V_2}{2}$$

$$A_{dm} = \frac{V_{01} - V_{02}}{V_1 - V_2}$$

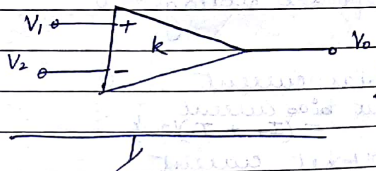
$$A_{cm} = \frac{V_{01} + V_{02}}{V_1 + V_2}$$

In case of single output, V_1



$$V_0 = A_d (V_1 - V_2) + A_c \frac{V_1 + V_2}{2}$$

Virtual Short or Virtual Ground:



$$V_0 = k(V_1 - V_2)$$

for 741

(*)

one pole
model of
op-amp

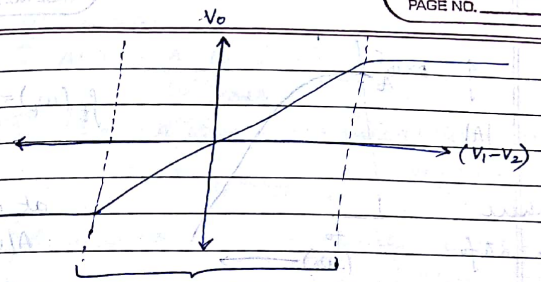
Different models to represent open-loop gain

(i) $A_0 = \infty$

↳ dc value of open loop gain.

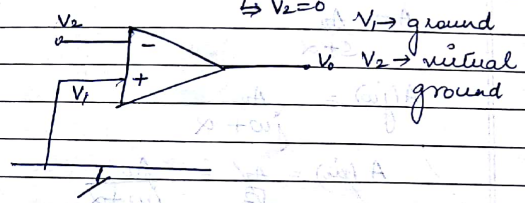
(ii) $A = \frac{A_0}{s + \alpha}$

(iii) $A = \frac{A_0 k}{\beta}$



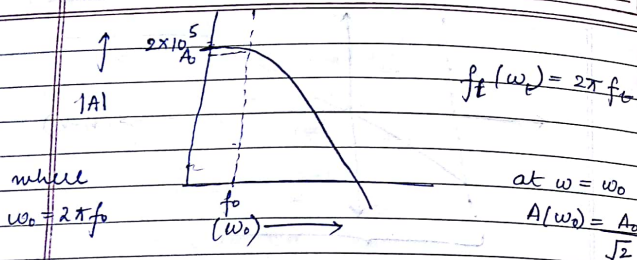
if, $k \rightarrow \infty \Rightarrow V_1 - V_2 = 0$

if $V_1 = 0$
 $\Rightarrow V_2 = 0$



(ve) feedback helps in maintaining linear region of operation

(+ve) feedback $\Rightarrow$ saturation.



$$A = \frac{A_0}{s + \alpha}$$

$$A(j\omega) = \frac{A_0}{j\omega + \alpha}$$

$$A(j\omega_0) = \frac{A_0}{\sqrt{2}} = \frac{A_0}{\sqrt{j\omega_0 + \alpha}}$$

$$\Rightarrow \alpha = \sqrt{2} - j\omega_0$$

$$\Rightarrow A(\alpha) = \frac{A_0}{\sqrt{2}\alpha} \text{ and } A(0) = \frac{A_0}{\alpha}$$

- i) dc gain
- ii) 3dB frequency
- iii) unity gain frequency

① $A_0 = \infty$

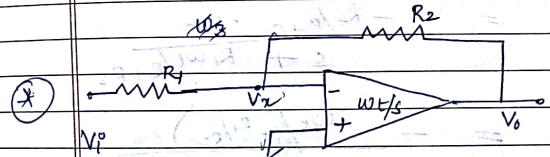
② $A(s) = \frac{A_0 \omega_{3dB}}{s + \omega_{3dB}}$

③ $A = \frac{A_0 \omega_{3dB}}{s = \omega_t/s} \rightarrow \text{integrator model of op-amp.}$

$\{ A_0 \omega_{3dB} = \omega_t \rightarrow \text{unity gain bandwidth.} \}$

if $\omega \gg \omega_{3dB}$

$$A(j\omega) = \frac{\omega_t}{\omega_{3dB} + j\omega}$$



$$V_i - V_x + \frac{V_o - V_x}{R_2} = 0$$

$$\text{as } V_x = -\frac{V_o}{\omega_t/s}$$

$$\frac{V_i + sV_o/\omega_t}{R_1} + \frac{V_o + sV_o/\omega_t}{R_2} = 0$$

$$V_i R_2 + \left(\frac{sR_2 + R_1 + R_1 s}{\omega_t} \right) V_o = 0$$

$$\Rightarrow \frac{V_o}{V_i} = \frac{-R_2}{s/\omega_t (R_1 + R_2) + R_1}$$

$$A = \frac{V_o}{V_i} = - \frac{R_2}{(R_1 + R_2) \frac{s}{\omega_c} + R_1}$$

$$= - \frac{R_2 / R_1}{(1 + \frac{R_2}{R_1}) \frac{s}{\omega_c} + 1}$$

pole, $\frac{R_1 + R_2}{R_1} \frac{s}{\omega_c} = -1$

$$\boxed{s = - \frac{\omega_c R_1}{R_1 + R_2}}$$

$$A = \frac{-R_2 \omega_c}{(R_1 + R_2) s + R_1 \omega_c}$$

$$= - \frac{R_2 / R_1 \cdot \omega_c}{s + R_1 \omega_c / (R_1 + R_2)}$$

$$A = - \frac{\omega_c / (1 + R_1 / R_2)}{s + \omega_c / (1 + R_2 / R_1)}$$

for positive feedback,

$$V_i - \frac{V_o s}{\omega_c} + V_o - \frac{s V_o}{\omega_c} = 0$$

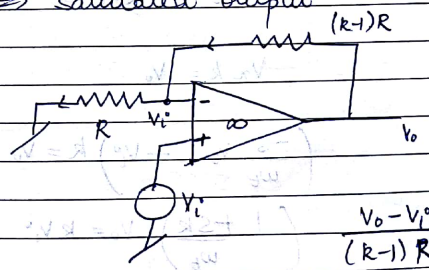
$$R_2 V_i - \frac{s R_2}{\omega_c} V_o + R_1 V_o - \frac{s R_1}{\omega_c} V_o = 0$$

$$R_2 V_i = V_o \left\{ \frac{s(R_2 + R_1)}{\omega_c} - R_1 \right\}$$

$$A = \frac{V_o}{V_i} = \frac{R_2 / R_1}{\frac{s}{\omega_c} (1 + R_2 / R_1) - 1}$$

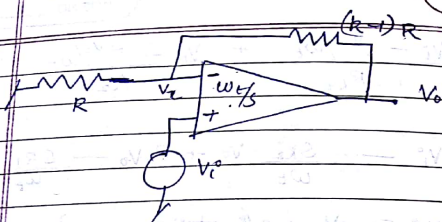
$$\Rightarrow A = \frac{R_2 / R_1 \cdot \omega_c / (1 + R_2 / R_1)}{s - \omega_c / (1 + R_2 / R_1)}$$

$\Rightarrow$ pole in RHS of s-plane
 $\Rightarrow$ unstable amplifier
 $\Rightarrow$ saturated output



$$\frac{V_o - V_i}{(R - 1) R} = \frac{V_i}{R}$$

$$\Rightarrow \boxed{\frac{V_o}{V_i} = R}$$



$$\frac{V_o}{V_i - V_x} = \frac{w_E}{s} \Rightarrow -sV_o + w_E V_i = V_x w_E$$

$$\frac{V_x - V_o}{(k-1)R} + \frac{V_x}{R} = 0$$

$$\frac{V_x}{(k-1)R} + \frac{V_x}{R} = \frac{V_o}{(k-1)R}$$

$$V_x \left[\frac{1+k-1}{k-1} \right] = \frac{V_o}{k-1}$$

$$V_x k = V_o$$

$$\left(\frac{-sV_o}{w_E} + V_i \right) k = V_o$$

$$\left(1 + \frac{s k}{w_E} \right) V_o = k V_i$$

$$\frac{A_o}{s + \frac{w_E}{k}}$$

$$\frac{V_o}{V_i} = \frac{k}{1 + s k / w_E}$$

$$= \frac{k w_E}{w_E + s k}$$

$$\Rightarrow A = \frac{V_o}{V_i} = \frac{w_E}{s + w_E/k}$$

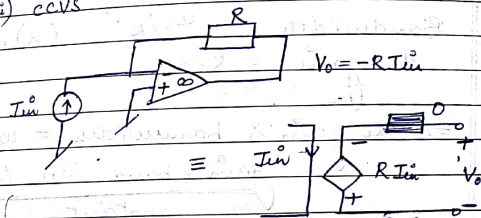
Bandwidth = w_E/k (α)
dc gain = k.

⇒ dc gain × bandwidth = w_E
gain × bandwidth is constant

(*) Realization of
(i) VCVS $\begin{cases} \text{non-inverting} \\ \text{inverting} \end{cases}$

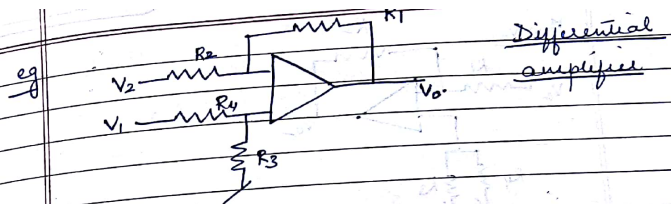
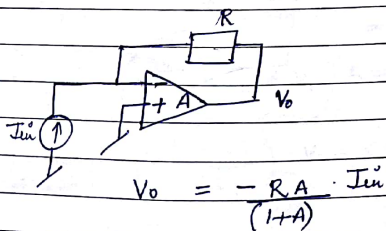
(ii) VCCS $\begin{cases} \text{for floating load} \\ \text{for grounded load} \end{cases}$

(iii) CCVS



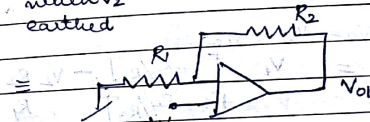
$$\text{Input impedance} = \frac{V_{in}}{I_{in}} = \frac{0}{I_{in}} = 0$$

Output impedance = 0.



Differential amplifier

$V = V_{o1} + V_{o2}$
 $\downarrow$ when V_1 earthed
 $\downarrow$ when V_2 earthed

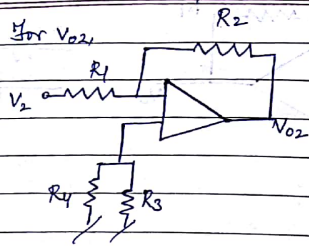


$$V' = \frac{R_3}{R_3 + R_4} V_1$$

$$V_{o1} = V' \left[1 + \frac{R_2}{R_1} \right]$$

$$= V_1 \cdot \frac{R_3}{R_3 + R_4} \left(1 + \frac{R_2}{R_1} \right)$$

$$= V_1 \cdot \frac{(1 + R_2/R_1)}{(1 + R_4/R_3)}$$



$$V_{02} = -\frac{R_2}{R_1} V_2$$

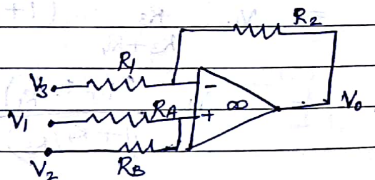
$$\text{Thus, } V_0 = V_{01} + V_{02} \\ = V_1 \left[\frac{1+R_2/R_1}{1+R_4/R_3} \right] - V_2 \frac{R_2}{R_1}$$

$$\text{If } \frac{R_4}{R_3} = \frac{R_1}{R_2}$$

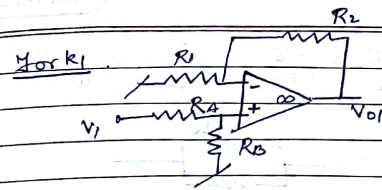
$$\text{Then, } V_{out} = (V_1 - V_2) R$$

or otherwise

$$V_{out} = R_1 V_1 - R_2 V_2$$



Differential Amplifier

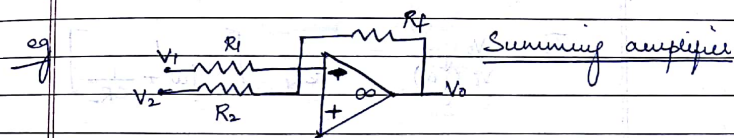


$$\Rightarrow K_1 = \frac{1 + R_2/R_1}{1 + R_4/R_3}$$

$$K_2 = \frac{1 + R_2/R_1}{1 + R_4/R_3}$$

$$K_3 = \frac{R_2}{R_1}$$

$$V_{out} = K_1 V_1 + K_2 V_2 - K_3 V_3$$

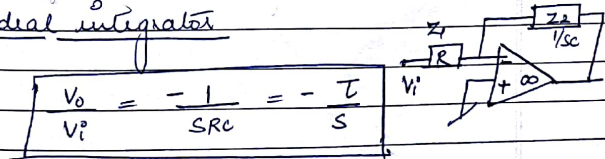


Summing amplifier

$$\text{Scaled summer} = -(K_1 V_1 + K_2 V_2)$$

$$= -\left(V_1 \frac{R_f}{R_1} + V_2 \frac{R_f}{R_2} \right)$$

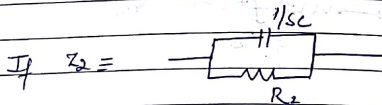
(*) Ideal integrator



$$\frac{V_0}{V_i} = \frac{-1}{sRC} = -\frac{1}{s}$$

{ RC circuits can't give ideal integrators, $\frac{V_0}{V_i} = \frac{1}{1+sRC}$ }

$$\frac{V_o}{V_i} = \frac{-I}{s} = \frac{-I}{j\omega} = \frac{jI}{\omega} = \frac{I}{\omega} \angle 90^\circ$$



$$\Rightarrow \frac{-Z_2}{Z_1} = \frac{-Y_1}{Y_2}$$

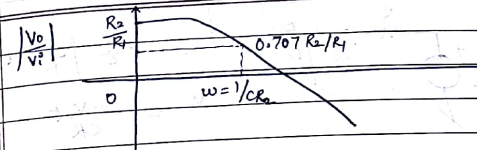
$$= - \frac{1/R_1}{1/R_2 + sC}$$

$$\Rightarrow \frac{V_o(s)}{V_i(s)} = - \frac{1}{R_1 C} \left[\frac{R_2}{s + 1/R_2 C} \right]$$

$$\text{If } \frac{V_o(j\omega)}{V_i(j\omega)} = - \frac{1}{CR_1} \left[\frac{1}{j\omega + \frac{1}{CR_2}} \right]$$

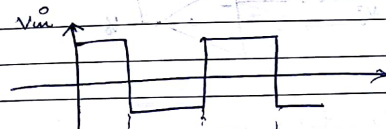
$$\text{If } \omega \gg 1/CR_2$$

$$\text{Then, } \frac{V_o(j\omega)}{V_i(j\omega)} = - \frac{1}{CR_1 j\omega}$$

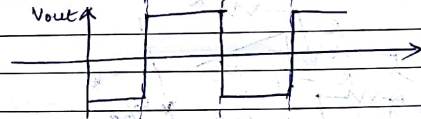


$$\left| \frac{V_o(j\omega)}{V_i(j\omega)} \right| = \frac{1}{R_1 C} \times \frac{1}{\sqrt{\omega^2 + (1/CR_2)^2}}$$

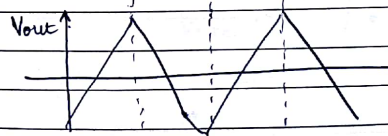
$$= \frac{R_2}{\sqrt{2} R_1} \quad (\text{at } \omega = 1/CR_2)$$

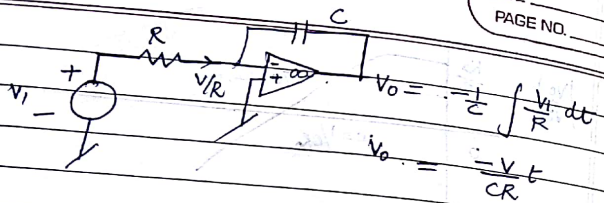


For $\omega \ll 1/CR_2$

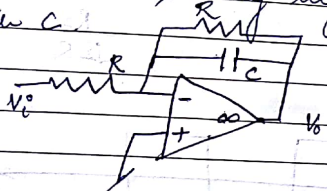


For $\omega \gg 1/CR_2$

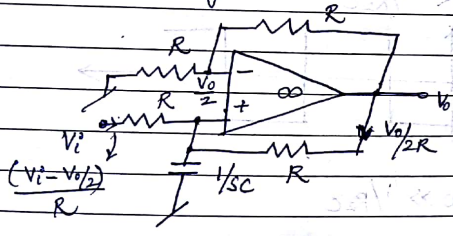




If i/p has dc component, ideal integrator can't be used coz of saturated output. To overcome this, very high R used in parallel with C.



(*) Miller's Integrator

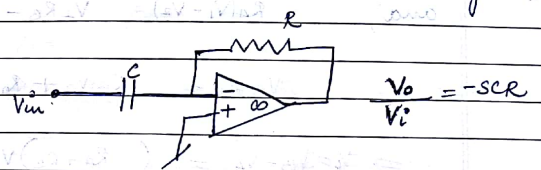


Using KCL.

$$\frac{V_i - V_o/2}{R} + \frac{V_o}{2R} = \frac{V_o}{sC}$$

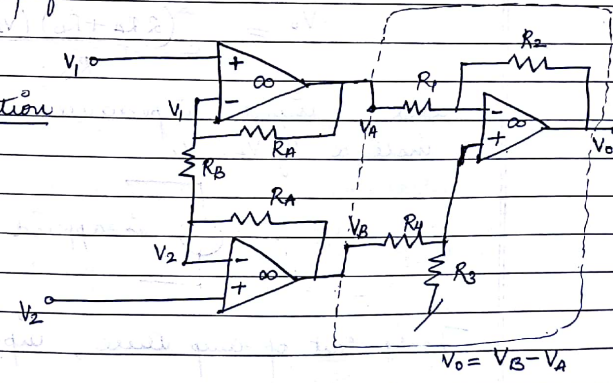
$$\Rightarrow \frac{V_o}{V_i} = \frac{2}{sRC}$$

non-inverting ideal integrator (Deboo's Integrator)



This circuit not used coz noise is also amplified.

Instrumentation Amplifier



$$A_0 \quad \frac{V_A - V_1}{R_A} = \frac{V_1 - V_2}{R_B} = \frac{V_2 - V_B}{R_A}$$

$$V_A R_B - R_B V_1 = R_A (V_1 - V_2)$$

$$\Rightarrow V_A = \frac{(R_A + R_B) V_1 - R_A V_2}{R_B}$$

and, $R_A (V_1 - V_2) = V_2 R_B - V_B R_B$

$$V_B R_B = R_B V_2 + R_A V_2 - R_A V_1$$

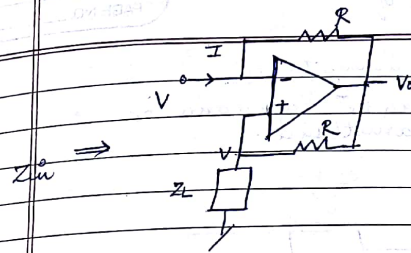
$$\Rightarrow V_B = V_2 - V_A = \frac{(R_A + R_B) V_2 - R_A V_1 - (R_A + R_B) V_1}{R_B} + R_A V_2$$

$$V_0 = \frac{(2R_A + R_B)(V_2 - V_1)}{R_B}$$

and input impedance = ∞ if instead of V_2, V_1 ,

$V_2 - V_1$ is applied.

If only last op-amp there, input impedance $\neq 0$.



Negative Impedance Converter (NIC)

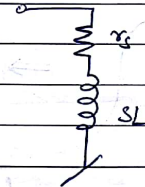
$$Z_{in}^0 = V/I$$

$$= -R \parallel Z_L$$

if $Z_L = sL$

practically, inductor is not lossless.

$r_s \rightarrow$ parasitic resistance of inductor



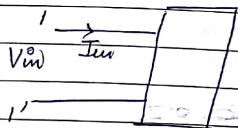
negative impedance converter $\rightarrow$

- $\rightarrow$ eliminate the effect of parasitic components.
- $\rightarrow$ design of active filters

creating effect of.

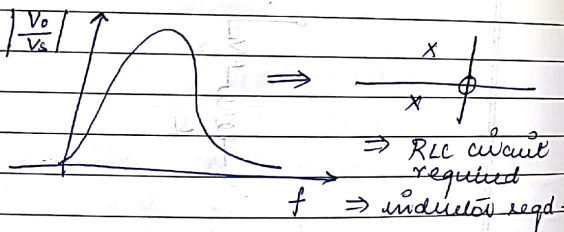
simulated inductance.

can be impedance or admittance.

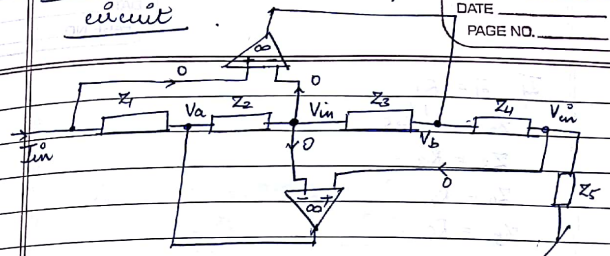


$$X_{in} = \frac{V_{in}}{I_{in}} (s) = sL_{eq}$$

want the effect of inductor without actually using an inductor.
 & coz of space considerations,



GIC - Generalized Impedance Converter circuit



$$I_{in} = \frac{V_{in} - V_a}{Z_1} \quad (1)$$

$$\frac{V_a - V_{in}}{Z_2} = -\frac{V_b + V_{in}}{Z_3} \quad (2)$$

$$\frac{V_{in}}{Z_5} = \frac{V_b - V_{in}}{Z_4} \quad (3)$$

$$V_{in} - I_{in} Z_1 = V_a \quad (a)$$

$$\frac{V_{in} Z_4 + V_{in}}{Z_5} = V_b \quad (b)$$

$$V_{in} [Z_4 + Z_5] = V_b Z_5$$

$$\Rightarrow Z_3 V_a - Z_3 V_{in} = Z_2 V_{in} - Z_2 V_b$$

$$V_{in} (Z_2 + Z_3) = Z_3 V_{in} - Z_2 I_{in} Z_1 +$$

$$\Rightarrow X_{in} = \frac{Z_1 Z_3 Z_5}{Z_2 Z_4}$$

If $Z_1 = R_1$

$Z_2 = 1/C_2 S$

$Z_3 = R_3$

$Z_4 = R_4$

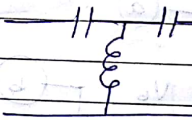
$Z_5 = R_5$

$Z_{in} = \frac{R_1 R_3 R_5 C_2 S}{R_4}$

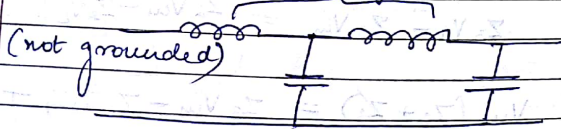
{ comparing with $Z = L_{eq} S$ }

$\Rightarrow L_{eq} = \frac{R_1 R_3 R_5 C_2}{R_4}$

Grounded Inductor



cannot create floating inductors using this



If $Z_2 = 1/C_2 S$

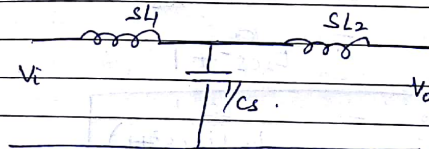
$Z_4 = 1/C_4 S$

$Z_{in} = R_1 R_3 R_5 C_2 C_4 S^2$

$= -\omega^2 k \Rightarrow$ frequency dependent (ve) resistor. (FDNR)

If $Z_1 = 1/C_1 S, Z_3 = 1/C_3 S, Z_2, Z_4, Z_5$

$\Rightarrow Z_{in}(s) = -k/\omega^2$

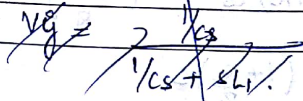


$V_i \neq S L_1 I_1(s) + (I_1 - I_2) \frac{1}{C S}$

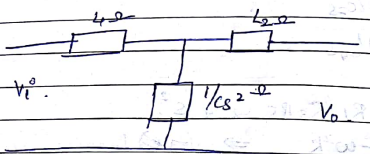
$V_i = I_1 (S L_1 + \frac{1}{C S}) - \frac{I_2}{C S}$

$V_o = \frac{I_2 - I_1}{C S} + I_2 S L_2$

$V_o = I_2 (\frac{1}{C S} + S L_2) - \frac{I_1}{C S}$



$\frac{V_o}{V_i} = \frac{S L_1 / (S L_2 + \frac{1}{C S})}{S L_1 + \frac{1}{C S} - \frac{I_2}{C S I_1}}$
 $= \frac{L_2 / C \times S C}{S^2 L_2 C + 1}$
 $= \frac{1}{S^2 L_2 C + 1}$

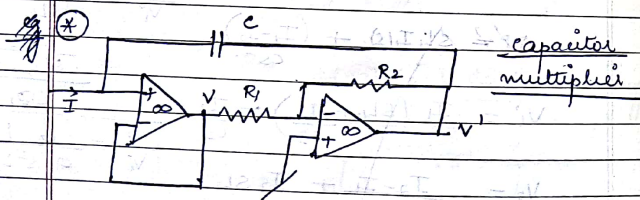


$$\frac{V_o}{V_i} = \frac{Z}{L_1 + Z}$$

$$Z = \frac{1/cs^2}{L_2 + 1/cs^2}$$

$$= \frac{1}{L_2 cs^2 + 1}$$

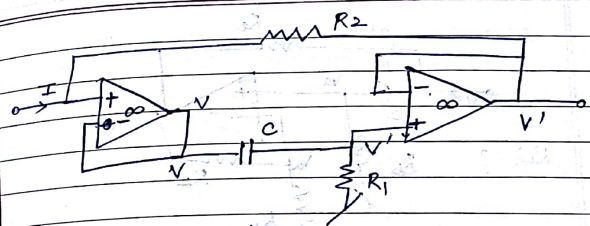
$$\frac{V_o}{V_i} = \frac{-K_2}{K_1 (L_2 cs^2 + 1)}$$



$$\frac{V}{I} = \frac{R_1}{(R_1 + R_2)cs} = \frac{1}{(1 + \frac{R_2}{R_1})cs}$$

Comparing with $1/cs$

$$\Rightarrow C_{eq} = \left(1 + \frac{R_2}{R_1}\right)C$$



$$V' = \frac{R_1 V}{R_1 + \frac{1}{cs}}$$

$$= \frac{R_1 cs V}{R_1 cs + 1}$$

$$I = \frac{V - V'}{R_2}$$

$$= \frac{V}{R_2} \left(\frac{R_1 cs + 1 - R_1 cs}{R_1 cs + 1} \right)$$

$$= \frac{V}{R_2 (R_1 cs + 1)}$$

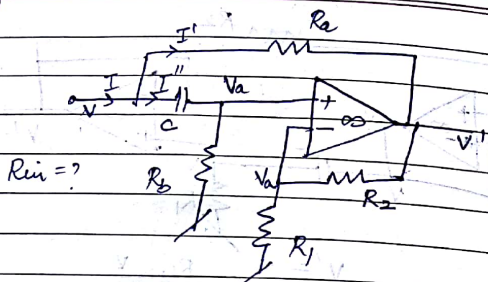
$$\Rightarrow \frac{V}{I} = R_2 + sCR_1R_2$$

$$\text{quality factor} = \frac{\omega L}{R} \approx \frac{(cR_1R_2)s}{R_2}$$

$$= \frac{\omega CR_1R_2}{R_2}$$

$$= \omega CR_1$$

lossy inductor is of series R-L type



$$I = I_1 + I_2$$

$$I' = \frac{V' - V}{R_a}$$

$$V_a = \frac{R_b}{R_b + \frac{1}{Cs}} V' = \frac{R_1}{R_1 + R_2} V'$$

$$\Rightarrow V' = \frac{(R_1 + R_2) R_b}{R_1 (R_b + 1/Cs)}$$

$$I' = \frac{V}{R_a} \left[\frac{(R_1 + R_2) R_b - R_1 R_b - R_1 / Cs}{R_1 (R_b + 1/Cs)} \right]$$

$$\frac{V}{I} = \frac{R_a R_1 (R_b + 1/Cs)}{R_2 R_b - R_1 / Cs}$$

$$= \frac{R_a R_1 (R_b Cs + 1)}{R_2 R_b Cs - R_1}$$

$$I = \frac{V_{in} sC}{1 + sCR_b} + \frac{1}{R_a} \left[V_{in} - \frac{(1 + R_2) sCR_b}{R_1} V_{in} \right]$$

$$\frac{I}{V_{in}} = \frac{sC}{1 + sCR_b} + \frac{1}{R_a} \left[1 - \frac{sCR_b}{R_1} (1 + R_2) \right]$$

$$= V_{in} \left[\frac{sC}{1 + sCR_a} + \frac{1}{R_a} - \frac{sCR_b}{1 + sCR_a} \left(\frac{2}{R_1} \right) \right]$$

$$= V_{in} \left[\frac{sC(1 - 2R_b/R_1) + 1}{1 + sCR_a} \right]$$

$$I = V_{in} \frac{R_a sC(1 - 2R_b/R_1) + 1 + sCR_a}{(1 + sCR_a) R_a}$$

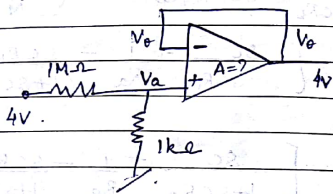
$$\text{If } R_1 = R_2, R_a = R_b$$

$$\frac{V_{in}}{I} = \frac{(1 + sCR_a) R_a}{2R_a sC / 2R_a Cs}$$

$$I = V_{in} \left[\frac{R_a sC - sCR_b (1 + R_2/R_1) + 1}{(1 + sCR_b) R_a} \right]$$

$$\frac{V_{in}}{I} = \frac{R_a sC - sCR_b - sCR_2 R_b / R_1 + 1 + sCR_b}{(1 + sCR_b) R_a}$$

$$\Rightarrow \frac{V}{I} = \frac{R_a R_1 [1 + sCR_b]}{R_1 + sC [R_a R_1 - R_2 R_b]} = R_{in}$$



$$V_a = \frac{1000 - 4V}{1000 + 1000}$$

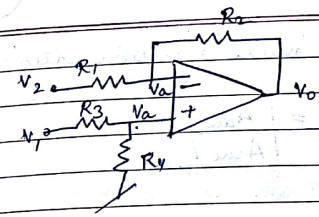
$$V_a = \frac{1}{1001} \times 4 = \frac{4}{1001}$$

$$V_o = A$$

$$V_a - V_a$$

$$A = - \frac{4 - \frac{4}{1001}}{1 - \frac{1}{1001}} = - \frac{1}{1000}$$

$$\Rightarrow A = 1.001$$



$$V_a = \frac{R_4 V_1}{R_3 + R_4}$$

$$\frac{V_a - V_2}{R_1} = \frac{V_o - V_a}{R_2}$$

$$R_2 V_a - R_2 V_2 = R_1 V_o - R_1 V_a$$

$$(R_1 + R_2) V_a = R_2 V_2 + R_1 V_o$$

$$(R_1 + R_2) \frac{R_4 V_1}{R_3 + R_4} = R_2 V_2 + R_1 V_o$$

$$R_1 V_o = \frac{(R_1 R_4 + R_2 R_4) V_1}{(R_3 + R_4)} - R_2 V_2$$

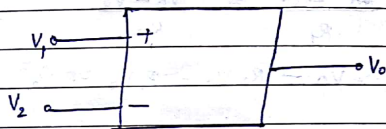
$$\Rightarrow V_o = \frac{R_4 (R_1 + R_2) V_1}{R_1 (R_3 + R_4)} - \frac{R_2 V_2}{R_1}$$

Common mode rejection ratio (CMRR)

$$CMRR = \left| \frac{A_{dm}}{A_{cm}} \right|$$

For A_{cm} , $V_1 = V_2 = V$

$$A_{cm} = \frac{V_o}{V_i} = \frac{1 + R_2/R_1}{1 + R_3/R_4} - \frac{R_2}{R_1}$$



eg If common mode gain = 10 dB
and diff. mode gain = 120 dB.

$$\Rightarrow A_{cm} = 20 \log_{10} 10$$

$$A_{cm} = 20 \log_{10} A_c = 10$$

$$A_c = 3.16$$

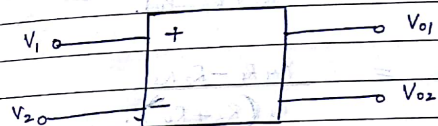
$$A_d = 10^6$$

$$CMRR = \frac{A_d}{A_c} = \frac{10^6}{3.16} = 3.16 \times 10^5$$

$$V_{id} = V_1 - V_2$$

$$V_{icm} = \frac{V_1 + V_2}{2}$$

$$V_o = R_1 V_1 - R_2 V_2 = R_3 V_{icm} + R_4 V_{id}$$



$$V_{id} = V_1 - V_2$$

$$V_{od} = V_{o1} - V_{o2}$$

$$V_{icm} = \frac{V_1 + V_2}{2}$$

$$V_{ocm} = \frac{V_{o1} + V_{o2}}{2}$$

$$V_{o1} - V_{o2} = R_1 V_{id} + R_2 V_{icm}$$

eg
-
Adm = ?
Adm = ?

$$A_{dm} = \frac{R_4 R_1 + R_2 R_4 - R_2}{R_1 R_3 + R_4 R_1 - R_1}$$

$$= \frac{R_4 R_1 + R_2 R_4 - R_2 R_3 - R_1 R_4}{R_1 (R_3 + R_4)}$$

$$A_{dm} = \frac{R_4 R_1 - R_2 R_3}{R_1 (R_3 + R_4)}$$

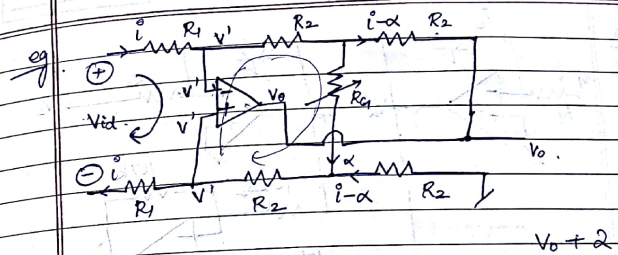
$$V_{id} = V_1 - V_2$$

$$V_{cm} = \frac{V_1 + V_2}{2}$$

$$V_1 = V_{cm} + \frac{V_{id}}{2}$$

$$V_2 = V_{cm} - \frac{V_{id}}{2}$$

$$A_{dm} = \frac{1}{2} \left(\frac{1 + R_2/R_1}{1 + R_3/R_4} + \frac{R_2}{R_1} \right)$$



$$i = \frac{V_{id}}{2R_1}$$

$$i R_2 + 0 + i R_2 + V_{ce} = 0$$

$$+ \alpha R_E = V_{ce} = -2i R_2$$

$$= -\frac{2 V_{id} R_2}{2 R_1} = -\frac{V_{id} R_2}{R_1}$$

$$V_{ce} = -2(i - \alpha) R_2 + V_{ce}$$

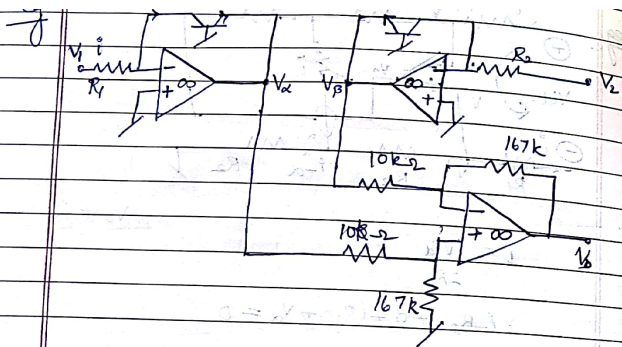
$$i - \alpha = \frac{V_{id}}{2 R_1} + \frac{V_{id} R_2}{R_1 R_E}$$

$$= \frac{V_{id}}{R_1} \left[\frac{1}{2} + \frac{R_2}{R_E} \right]$$

$$V_{ce} = -\frac{V_{id}}{R_1} \left[R_2 + \frac{R_2^2}{R_E} \right] - \frac{V_{id} R_2}{R_1}$$

$$= -\frac{V_{id}}{R_1} \left[R_2 + \frac{R_2^2}{R_E} + R_2 \right]$$

$$V_{ce} = -\frac{V_{id}}{R_1} \left[\frac{R_2^2}{R_E} + 2 R_2 \right] \quad V_{ce} = \frac{V_{id}}{R_1} \left(\frac{R_2^2 + 2 R_2 R_E}{R_E} \right)$$



$$V_0 = \frac{167}{10} (V_x - V_p)$$

$$i = \frac{V_1}{R_1}$$

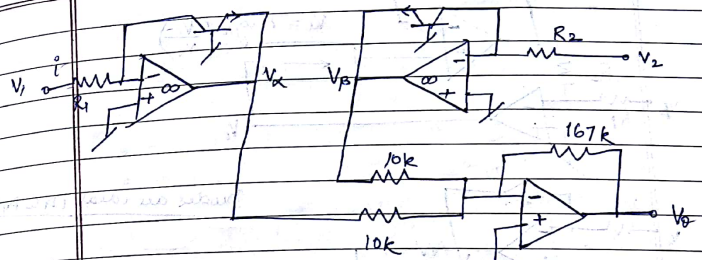
$$I_c = I_s e^{V_{BE}/V_T}$$

$$V_x = -V_T \ln \frac{I_c}{I_s}$$

$$V_0 = \frac{167}{10} \left(-V_T \ln \frac{V_1}{R_1 I_s} + V_T \ln \frac{V_2}{R_2 I_s} \right)$$

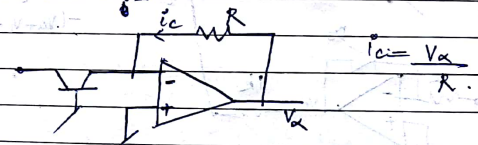
$$= \frac{167}{10} \left(V_T \ln \frac{V_2 R_1}{R_2 V_1} \right)$$

$$= \ln \frac{V_2 R_1}{V_1 R_2} \text{ Volts}$$



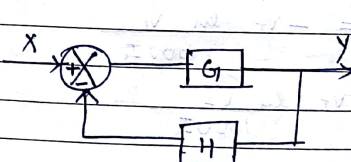
$$V_x + V_p = -V_T \ln \frac{V_1 V_2}{I_s^2 R_1 R_2}$$

$$V_0 = -16.7 [V_x + V_p]$$



$$I_c = \frac{V_x}{R}$$

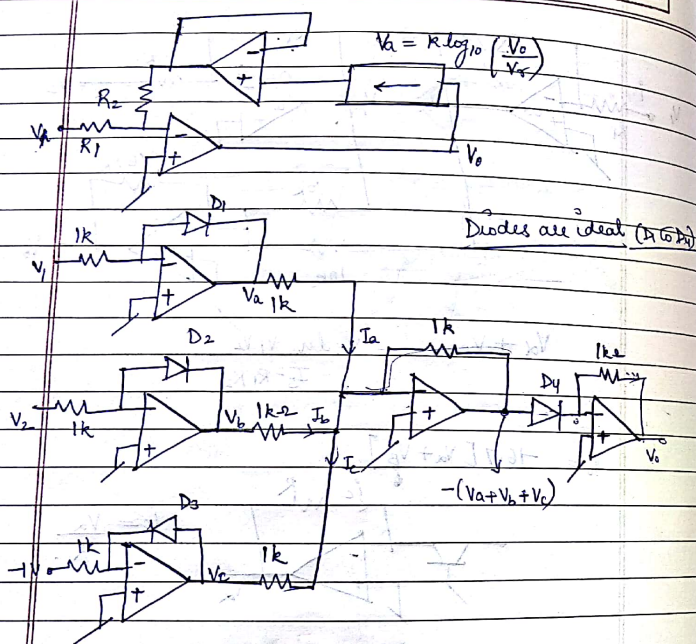
$$\frac{V_x}{R} = I_c = I_s e^{-V_1/V_T}$$



$$\frac{Y}{X} = \frac{G}{1 + GH}$$

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$$\frac{V_1}{1000} = I_s e^{-V_a/V_T}$$

$$\Rightarrow V_a = -V_T \ln \frac{V_1}{1000 I_s}$$

$$V_b = -V_T \ln \frac{V_2}{1000 I_s}$$

$$V_c = V_T \ln -1 = -V_T \ln 1000 I_s$$

$$V_0 = -V_1 V_2$$

$$V_a + V_b + V_c = -V_T \ln \frac{V_1 V_2}{1000 I_s}$$

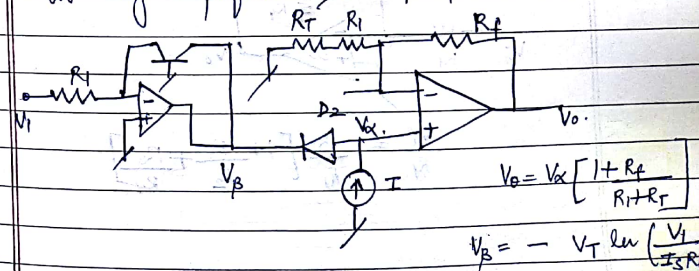
$$I_{D4} = I_s e^{+V_b/V_T}$$

$$= I_s \frac{1000 I_s}{V_2} \times \frac{V_1 V_2}{1000 I_s}$$

$$= \frac{V_1 V_2}{1000}$$

$$V_0 = -V_1 V_2$$

To maximise the temp dependence of output in log amplifier → temp dependent resistance.

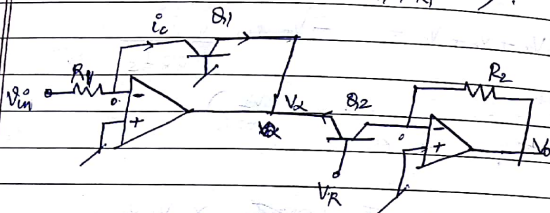


$$V_0 = V_x \left[\frac{1 + R_f}{R_1 + R_f} \right]$$

$$V_x = -V_T \ln \left(\frac{V_1}{I_s R_1} \right)$$

$$V_0 = V_T \ln \left(\frac{I R_T}{V_i} \right) \left\{ 1 + \frac{R_T}{R_T + R_1} \right\}$$

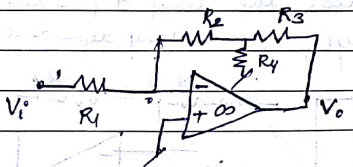
$$= V_T \ln \left(\frac{I R_T}{V_i} \right) \left(\frac{R_T + R_1 + R_T}{R_1 + R_T} \right)$$



$$V_0 = \frac{-R_2}{R_1} e^{V_R/V_T}$$

~~Min~~

$$- \frac{V_{in}}{R_1} I e^{V_R/V_T}$$



$$\frac{V_i}{R_1} = \frac{-R_2}{R_1} \left[1 + \frac{R_2}{R_2} + \frac{R_3}{R_4} \right]$$